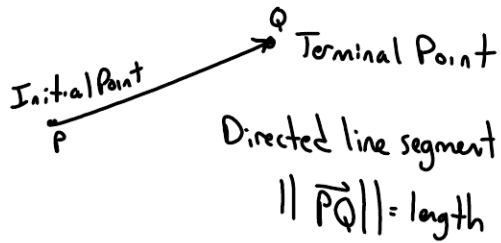


Ameet Rao's Calculus III Notes

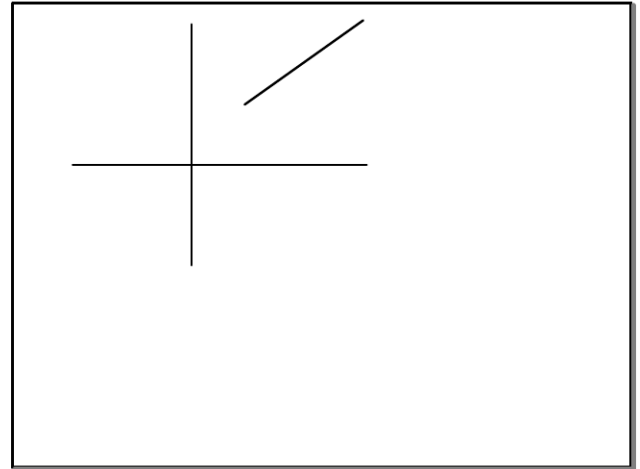
Vectors in 2D and 3D, Unit Vectors, Distance and Spheres, and the Dot Product



All directed line segments of the same length are equivalent

The set of all equivalent directed line segments is a vector

A vector is a whole family of arrows: every directed segment from an initial point P to a terminal point Q with the same length and direction. Its length is $\|\vec{PQ}\|$.



Sliding a vector somewhere else doesn't change it. Only its length and direction matter, not where it starts.

$$\vec{v} = \langle \Delta x, \Delta y \rangle$$

$$= \langle v_1, v_2 \rangle$$

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$$

A vector shows how far you move and in which direction. Use the horizontal and vertical changes as the legs of a right triangle: $\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$.

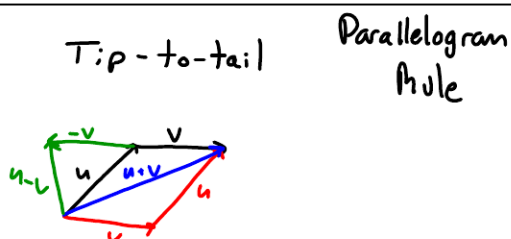
$$\vec{u} = \langle u_1, u_2 \rangle \quad \vec{v} = \langle v_1, v_2 \rangle$$

$$\vec{u} \pm \vec{v} = \langle u_1 \pm v_1, u_2 \pm v_2 \rangle$$

$$c\vec{v} = \langle cv_1, cv_2 \rangle$$

$$-\vec{v} = \langle -v_1, -v_2 \rangle$$

Add or subtract component by component. A scalar c stretches or shrinks the vector; a negative sign flips its direction.



Two ways to add: tip-to-tail (start the second where the first ends) or the parallelogram rule (share a starting point). Both land on the same diagonal, $\vec{u} + \vec{v}$.

$$\|c\vec{v}\| = |c| \|\vec{v}\|$$

Unit Vector - same direction
magnitude = 1

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\|\vec{v}\|} \vec{v}$$

Find the unit vector for $\langle 2, 5 \rangle$

$$\vec{u} = \left\langle \frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} \right\rangle = \frac{1}{\sqrt{29}} \langle 2, 5 \rangle$$

Scaling by c scales length by $|c|$. A unit vector has length 1 and the same direction: divide $\vec{v}$ by $\|\vec{v}\|$. Here $\|\langle 2, 5 \rangle\| = \sqrt{29}$, so $\vec{u} = \langle 2/\sqrt{29}, 5/\sqrt{29} \rangle$.

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Standard unit vector

$$\begin{aligned} i &= \langle 1, 0 \rangle & v &= \langle 3, -6 \rangle \\ j &= \langle 0, 1 \rangle & &= 3i - 6j \end{aligned}$$

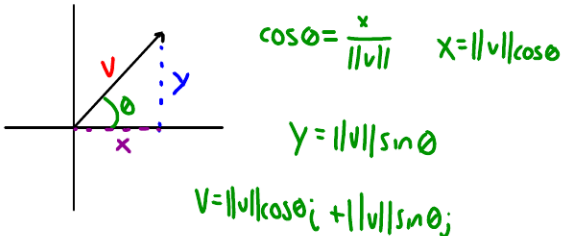
i and j are the building blocks: one step right, one step up. Every 2D vector is a mix of them, so $\langle 3, -6 \rangle = 3i - 6j$.

u has IP $(2, -5)$ TP $(1, -3)$

$$v = 2i - j \quad w = 2u - 3v$$

$$w = -8i + 7j$$

Components are terminal minus initial: $u = \langle 1-2, -3-(-5) \rangle = \langle -1, 2 \rangle$. Then combine: $2u - 3v = \langle -8, 7 \rangle = -8i + 7j$.

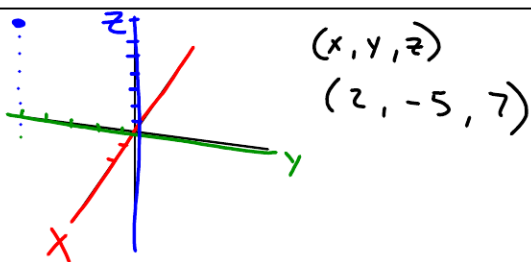


Trig turns length and angle into components: $x = \|v\| \cos \theta$, $y = \|v\| \sin \theta$. So $v = \|v\| \cos \theta i + \|v\| \sin \theta j$.

$$\|v\| = 3 \quad \text{Angle with } x \text{ axis is } \pi/6$$

$$v = \frac{3\sqrt{3}}{2} i + \frac{3}{2} j$$

Putting it to work: with $\|v\| = 3$ and $\theta = \pi/6$, $x = 3 \cos(\pi/6) = 3\sqrt{3}/2$ and $y = 3 \sin(\pi/6) = 3/2$.



3D adds a z -axis. Plot $(2, -5, 7)$ by going 2 along x , 5 in the negative y direction, then 7 up.

$$\begin{aligned} D &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \\ &\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right) \\ (x-h)^2 + (y-k)^2 + (z-l)^2 &= r^2 \end{aligned}$$

Distance: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$. Midpoint: average each coordinate. A sphere is every point at distance r from (h, k, l) : $(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$.

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$$\begin{aligned} i &= \langle 1, 0, 0 \rangle & \text{IP } (-2, 3, 1) \\ j &= \langle 0, 1, 0 \rangle & \text{TP } (0, -4, 4) \\ k &= \langle 0, 0, 1 \rangle & V = \langle 2, -7, 3 \rangle \\ & & \|v\| = \sqrt{62} \\ & & u = \frac{1}{\sqrt{62}} \langle 2, -7, 3 \rangle \end{aligned}$$

Same idea in 3D with i, j, k . Terminal minus initial gives $v = \langle 2, -7, 3 \rangle$; divide by $\|v\| = \sqrt{62}$ for the unit vector.

Dot Product

$$u = \langle u_1, u_2, u_3 \rangle$$

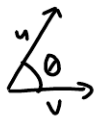
$$v = \langle v_1, v_2, v_3 \rangle$$

$$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3$$

$$u = \langle 2, -2, 5 \rangle \quad v = \langle 8, -4, 3 \rangle$$

$$u \cdot v = 39$$

The dot product multiplies matching components and adds them, giving a number, not a vector: $\langle 2, -2, 5 \rangle \cdot \langle 8, -4, 3 \rangle = 16 + 8 + 15 = 39$.



$$0 < \theta < \pi$$

$$\xrightarrow{u} \xrightarrow{v} \quad \theta = 0$$

$$\xleftarrow{u} \xrightarrow{v} \quad \theta = \pi$$

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|}$$

$$u \cdot v = \|u\| \|v\| \cos \theta$$

$$\text{if } u \cdot v = 0$$

u, v are
orthogonal

$$\theta = \frac{\pi}{2}$$

The dot product measures alignment: $\cos \theta = u \cdot v / (\|u\| \|v\|)$. Same direction gives $\theta = 0$, opposite gives $\theta = \pi$, and $u \cdot v = 0$ means perpendicular ($\theta = \pi/2$).

$$u = \langle 3, -1, 2 \rangle$$

$$v = \langle -4, 0, 2 \rangle$$

$$w = \langle 1, -1, -2 \rangle$$

$$z = \langle 2, 0, -1 \rangle$$

Find angle between

$$u, v \quad \theta = \cos^{-1} \left(\frac{u \cdot v}{\|u\| \|v\|} \right)$$

$$u, w \quad \theta = \frac{\pi}{2}$$

$$v, z \quad \theta = \pi$$

For each pair, find the dot product and both lengths, then take $\cos^{-1}$. Here $u \cdot w = 0$ (perpendicular) and $v \cdot z = -\|v\| \|z\|$, so v and z point opposite ways ($\theta = \pi$).